

1

C02 - AT3 - CLASS TEST

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24

25

6



Q. Explain the concept of image enhancement in the spatial domain.

### Image Enhancement:

Image enhancement is the process of improving the visual quality of an image or making it more suitable for analysis by human or computer. In spatial domain, enhancement is performed by directly modifying the intensity values of the pixels.

The spatial domain transformation is represented by

$$g(x, y) = T[f(x, y)]$$

where

$f(x, y) \rightarrow$  Input images.

$T \rightarrow$  Transformation function

$g(x, y) \rightarrow$  Enhanced image.

Techniques of Spatial Domain Enhancement

- \* Gray-level Transformation
- \* Histogram Processing
- \* Spatial filtering.

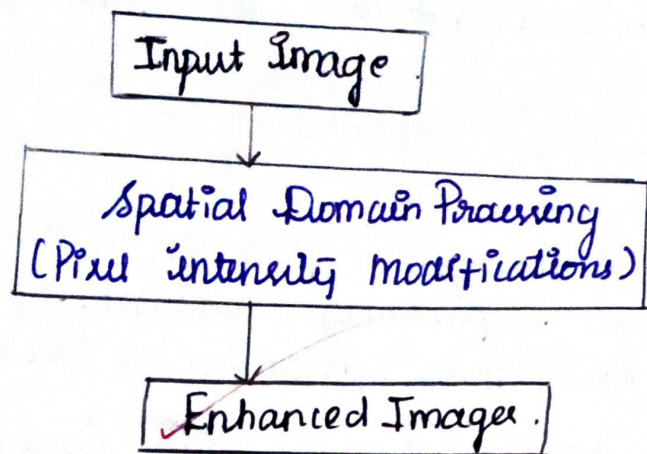
### Gray-level Transformation:-

These techniques modify the brightness and contrast of an image by changing the pixel intensity values.

example:

- $\rightarrow$  Image negative
- $\rightarrow$  Log transformation
- $\rightarrow$  Gamma (Power law)
- $\rightarrow$  Contrast stretching





## ② Differentiate between Image Smoothing & Image Sharpening.

Image enhancement techniques are used to improve the quality of digital images. Two important enhancement techniques are image smoothing & image sharpening. Image smoothing removes unwanted noise and reduces intensity variation, whereas image sharpening enhancement edges and fine details to make the image clear.

| Image Smoothing                    | Image Sharpening                   |
|------------------------------------|------------------------------------|
| Removes noise from an image        | Enhance edges and fine details     |
| Uses Low-Pass filter (LPF)         | Uses High-Pass filter (HPF)        |
| Produces a blurred or smooth image | Produces a sharper & clearer image |
| Reduces Intensity changes          | Increase intensity changes         |
| Before Image Segmentation          | After Image Smoothing              |



Uses: Improves dark or low contrast images.

## 2. Histogram Processing :-

A histogram represents the distribution of pixel intensity values in an image. Histogram equalization enhances the overall contrast by spreading the intensity values uniformly.

Uses: Medical images, satellite images.

## 3. Spatial Filtering :-

Spatial filtering processes each pixel using its neighboring pixels.

Types:

⇒ Smoothing Filter [Mean, Gaussian, Median]  
Remove noise and smooth the image.

⇒ Sharpening Filter [Laplacian, Sobel].  
Enhance edges & fine details.

### Advantages :-

- Improves image quality
- Reduces the noise
- Enhance edges & details.

### Application :-

- Medical imaging
- Remote sensing
- Face recognition

Spatial domain image enhancement directly modifies pixel intensity values to improve image quality (gray-level transformation adjust brightness, histogram processing improve contrast). Making the image more suitable for visualization & analysis.



## 1. Image Smoothing

Image Smoothing is the process of reducing noise and small intensity variation in an image.

### Working Principle:

- A filter mask (kernel) is moved across image
- pixel values are replaced by the average or median value.
- This reduces noise & smooths the image.

### Common Smoothing Filters:

- ★ Mean filter
- ★ Gaussian filter
- ★ Median filter.

## 2. Image Sharpening

Image Sharpening is the process of highlighting edges and fine details by emphasizing intensity changes in images.

### Working Principle:

- Detect rapid changes in pixel values.
- Enhance high-frequency component
- Make object boundaries more visible.

### Common Sharpening Filters:

- ★ Laplacian filter
- ★ Sobel filter
- ★ Prewitt Filter
- ★ High-Boost filter



13. Explain the Image Enhancement process in the frequency Domain.

Image enhancement in the frequency domain improve Image Quality by modifying the frequency components of an images instead of directly changing pixel values. The images is first transformed into the frequency domain using the Fourier transforms (FT) using the inverse Fourier Transforms (IFT).

$$G(u,v) = H(u,v) \times F(u,v)$$

Where

$F(u,v)$  = Input image frequency domain

$H(u,v)$  = Filter function

$G(u,v)$  = Enhanced image

Low - Pass Filter (LPF).

- ★ Allow low frequency components to pass.
- ★ Blocks high frequency components
- ★ Removes noise and smooth the images.

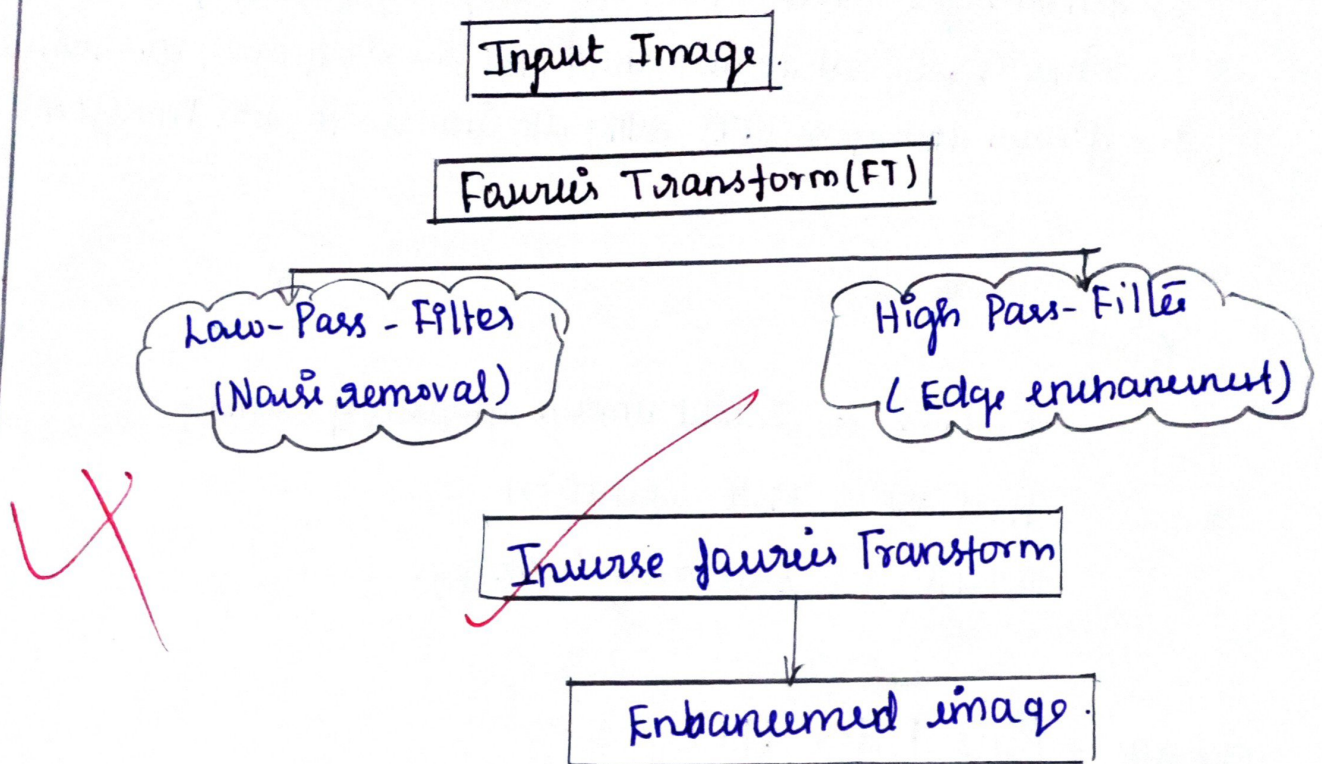
Application:-

- Medical Imaging
- Image denoising
- satellite image



## 2 High-Pass Filter (HPF)

- ★ Allows high-Frequency components to pass
- ★ Blocks low-Frequency components.
- ★ Enhance edges & fine details





Q4. Explain the concept of image segmentation.

Image Segmentation is the process of dividing an image into meaningful region or object. It simplifies images analysis and helps in object detection, recognitions and measurements.

### Segmentation Techniques

- \* Point Detection
- \* Line Detection
- \* Edge Detection
- \* Thresholding
- \* Region-Based Segmentation

Point detection :-

Detects isolated pixels whose neighbouring pixels are zero. Detection of dead pixel.

Line Detection :-

Detects horizontal, verticals, diagonal. Road extraction, document analysis.

Edge Detection :-

Detects object boundaries by sudden changes in pixel intensity. Face recognition.

Thresholding :-

Converts a grayscale image to a binary image. OCR.

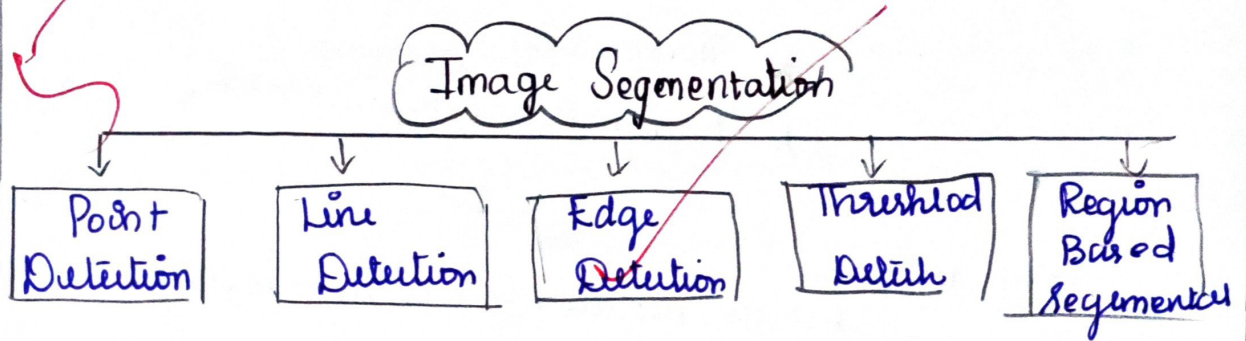
Region-Based Segmentation.

Group neighboring pixels with similar intensity.

MRI, CT Scan.



| Technique       | Advantages         | Limitations           | Application      |
|-----------------|--------------------|-----------------------|------------------|
| Point Detection | Isolated points    | Sensitive to noise    | Defect Detection |
| Line Detection  | Straight lines     | Orientation dependent | Road Detection   |
| Edge Detection  | Boundary detection | Noise sensitive       | Object Detection |





on the Importance of Edge Detection, edge linking.

Edge detection, edge linking and boundary detection are important techniques in image processing. They help identify object shapes, defects and improves object recognition by accurately location object boundaries.

### 1. Edge Detection:-

- ⊗ Detect sudden changes in pixel to identify object edges.
- ⊗ Common Operator: Sobel, Prewitt, Roberts.
- ⊗ Feature recognition - medical imaging, defect detection

### 2. Edge Linking:-

- ⊗ Connects broken or discontinuous edges to form continuous boundaries.
- ⊗ Uses pixel connectivity and gradient recognition

### 3. Boundary Detection:-

- ⊗ Identify the complete outer boundary of an object
- ⊗ Help measure object size & shape accurately.



## Importance

- Detects object boundaries accurately
- Improves object recognition
- Help identify cracks, holes, & defect.
- Increase the accuracy of image analysis

